MARK NYAMU

DESIGN, MANUFACTURING & MATERIALS ENGINEERING

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PROFILE

Chemical Engineering graduate with MSc study in Digital Design and Manufacturing, awaiting the final result. Practical project experience in CAD, additive manufacturing, polymer processing, materials testing and scan-to-CAD validation. Seeking graduate R&D, design engineering and product development opportunities, ideally in Manchester.

TECHNICAL SKILLS

Design: Autodesk Fusion 360, parametric modelling, T-Splines, generative design, Design for Additive Manufacturing (DfAM), Gravity Sketch.

Manufacturing & materials: LFAM, FFF/FDM, robotic pellet extrusion, recycled polypropylene feedstock, rapid prototyping, tensile and flexural testing, Granta EduPack.

Analysis & metrology: FEA, Creaform HandySCAN, Artec Leo, EinScan SP, scan registration, mesh processing, reverse engineering and CAD comparison.

SELECTED ENGINEERING PROJECTS

Recycled Polymer Feedstock for Large-Format Additive Manufacturing

MSc research | Manchester Metropolitan University | 2025-2026

- Investigated recycled polypropylene as feedstock for robotic pellet-extrusion LFAM, connecting material preparation, extrusion calibration, deposition trials and dimensional assessment.

- Prepared granulated feedstock and used controlled material-discharge measurements to characterise extrusion behaviour and support process calibration.

- Manufactured trial geometries using an ABB IRB 1600 robot with a Rapid Fusion Thermo-Prism twin-screw extrusion system.

- Captured specimens with a Creaform HandySCAN and compared registered scans with nominal CAD to assess geometric variation; distinguished dimensional evidence from qualitative observations of adhesion.

Aerospace Seat Bracket Optimisation

Generative design | FEA | Material testing | Additive manufacturing

- Reduced bracket mass from 163.85 g to 12.3 g, approximately 92.5%, through Fusion 360 generative design and manufacture in Markforged Onyx.

- Compared tensile and flexural behaviour of additively manufactured materials and defined mounting interfaces, obstacle geometry and 150 N / 200 N load cases for static stress analysis.

- Applied a design safety factor constraint of 2.0 and evaluated generated geometry through FEA and project testing.

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PRODUCT DEVELOPMENT & REVERSE ENGINEERING

Bespoke VR Headset Architecture

- Used Artec Leo scanning and photogrammetry to develop scan-derived head and hair geometry for a headset concept accommodating afro-textured hair.

- Combined Gravity Sketch ideation with Fusion 360 modelling and generative design, using contact regions and obstacle geometry to explore a hair-clearance structure.

Bike Curious: Urban Electric Moped Concept

- Developed an urban mobility concept through CAD and iterative FDM prototyping, exploring sustainable materials and refining geometry for manufacture.

- Split the model into pinned assemblies to reduce support requirements and refined print settings and fragile features in response to prototype failures.

Reverse Engineering: Aquarium Pump Lid

- Reconstructed a replacement lid from EinScan SP scan data using mesh repair, section sketches and CAD modelling in Fusion 360.

- Completed six prototype iterations to refine the snap-fit interface, incorporating ventilation and an organic upper-shell design.

INDEPENDENT DESIGN RESEARCH

Hunched Society | Footwear concept and prototyping

Founder & designer, independent concept project | January 2026-present

- Exploring a mono-material footwear concept combining wide-toe-box and zero-drop design principles with streetwear aesthetics.

- Using Fusion 360 for 3D form development and sole-unit prototyping; researching gyroid and Voronoi lattices as potential alternatives to conventional foams and adhesive-based construction.

PROFESSIONAL EXPERIENCE

Junior Marketing Executive | Sync

June 2024-November 2025

- Conducted market and competitor research using Google Trends, developed customer personas and produced SEO-focused web and blog content.

- Coordinated campaign briefs, brainstorming sessions and client acquisition events, building experience in communicating ideas and organising delivery.

EDUCATION

MSc Digital Design and Manufacturing | Manchester Metropolitan University

September 2025-September 2026 | Awaiting final result

Focus: additive manufacturing, DfAM, digital design, sustainable materials and manufacturing workflows.

BEng Chemical Engineering | Swansea University

2018-2021 | Second-class honours, lower division (2:2)

Foundation in polymer chemistry, material properties, thermodynamics and process control. Final capstone: large-scale chemical plant design, including reactors and safety instrumentation.